REGRESSION ANALYSIS

Measuring and predicting the relationship between variables

4.1 Mathematical and Statistical Equation – Meaning of Intercept and Slope
4.2 Error term – Measure for Model Fit –R2 – MAE – MAPE.
4.3 Multiple Linear Regression Model,
4.4 Partial Regression Coefficients,
4.5 Testing Significance overall significance of Overall fit of the model,
4.6 Testing for Individual Regression Coefficients

What Is Regression?

- **Regression** is a statistical technique used to model and analyze the relationships between variables. It helps in predicting the value of a dependent (target or explanatory) variable based on one or more independent (predictor) variables

- **Measuring relationships**

  Regression analysis measures the nature and extent of the relationship between two or more variables, enabling us to make predictions.

- **Average relationship**

  Regression is the measure of the average relationship between two or more variables.

Correlation vs. Regression

Basis	Correlation	Regression
Degree & Nature of Relationship	A measure of the degree of relationship between X and Y.	Studies the nature of the relationship so one may predict the value of one variable based on another.
Cause & Effect Relationship	Does not always assume a cause-and-effect relationship between two variables.	Clearly expresses the cause-and-effect relationship. The independent variable is the cause and the dependent variable is the effect.
Prediction	Does not help in making predictions.	Enables predictions using the regression line.
Symmetric	Coefficients are symmetrical: rxy = ryx.	Coefficients are not symmetrical: bxy ≠ byx.
Origin & Scale	Independent of the change of origin and scale.	Independent of change of origin, but not of scale.

Types of Regression Analysis

- **Simple & Multiple Linear Regression**

  Based on the number of independent variables used to predict the dependent variable.

- **Linear & Non-Linear Regression**

  Based on whether the relationship between variables is a straight line or a curve.

- **Partial & Total Regression**

  Based on whether all independent variables, or only some, are considered in the analysis.

Simple Regression Analysis

- **What is Simple Linear Regression**

  Simple linear regression is a statistical method used to model the relationship between two variables by fitting a linear equation to the observed data.

  One variable is considered the independent variable (predictor or explanatory variable), and the other is the dependent variable (response or outcome variable).

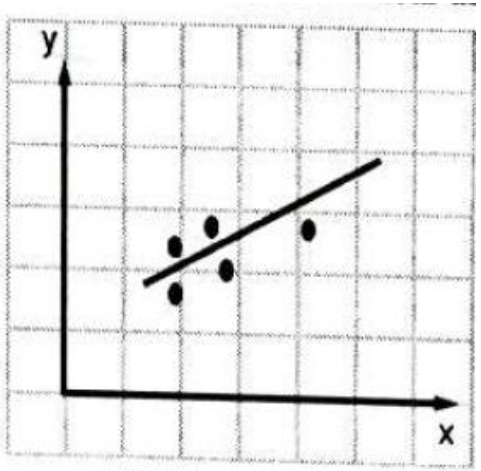
- **Key Components:**

  Independent Variable (X): The variable that is used to predict the value of the dependent variable.

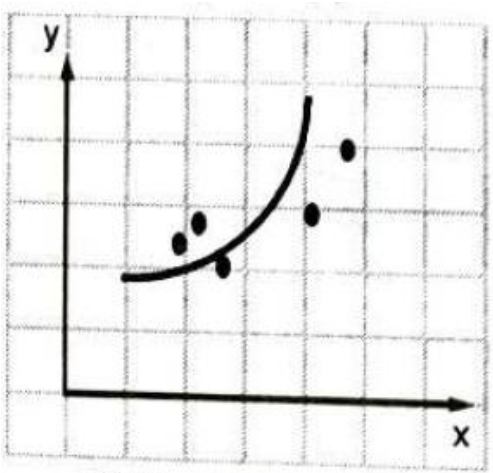
  Dependent Variable (Y): The variable that we are trying to predict or explain.

Simple Regression Analysis

- In simple regression training model records are plotted as a value of X vs. Value of Y
- Next task is to find a function so that if a random unknown value X is given, we can predict Y.
- There are different types of functions that can be used. In linear regression we assume that function is a linear function as shown in first graph



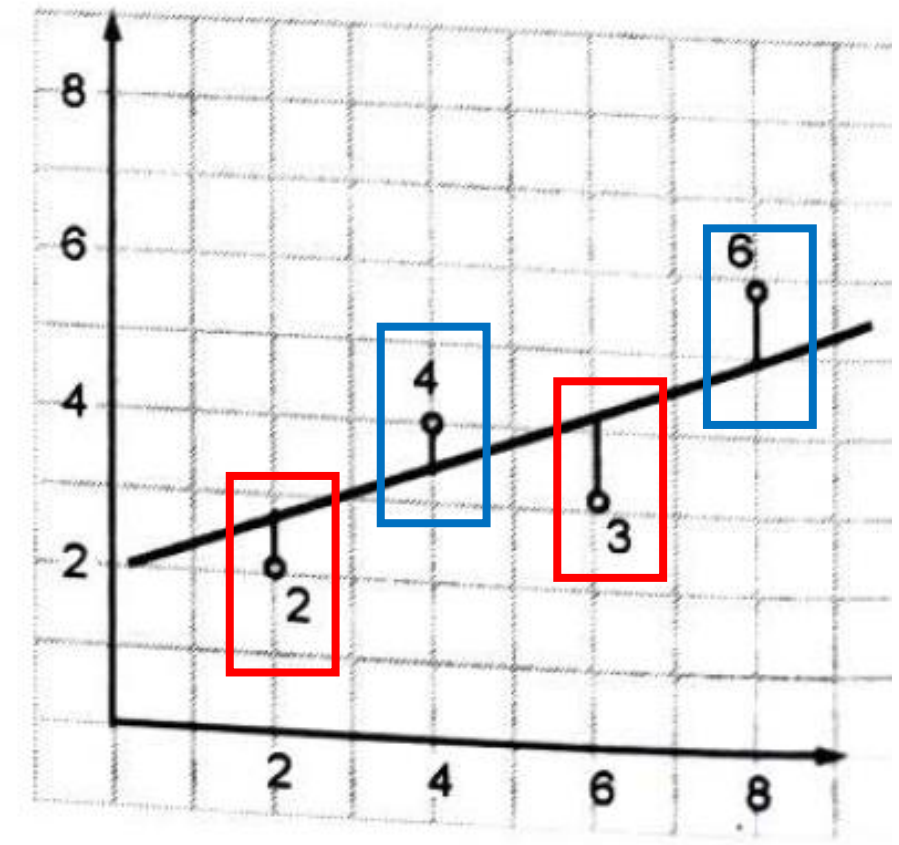
(a) Linear Function



(b) Non-Linear Function

Simple Regression Analysis

- In linear regression the best fitted straight line is drawn which passes through the points called regression line.
- Regression line shows calculated value of Y for each possible value of X.
- Errors in the prediction are shown by the vertical lines drawn from the points to the regression line
- Error in prediction is lees when point is very near to the regression line.
- When point is far from the regression line the error in the prediction is high;
- The difference between the value of a point and the predicted value is called error of the prediction.
- Predicted value is value of a point on the line.



Prediction and the Error in the Prediction

Simple Regression Analysis

- **Steps in Simple Linear Regression:**
  1. **Data Collection:** Gather data for the independent and dependent variables.
  2. **Calculate the Slope (a) and Intercept (b):**
     1. The slope a is calculated using the formula: $a = \frac{n(\Sigma xy)-(\Sigma x)(\Sigma y)}{n(\Sigma x^2)-(\Sigma x)^2}$ $\boxed{\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}}$
     2. The intercept b is calculated using the formula: $b = \frac{\Sigma y - b(\Sigma x)}{n}$ $\boxed{\bar{y}-a\bar{x}}$

        Where n the number of data points.
  3. **Fit the Regression Line:** Use the calculated slope and intercept to fit the best line through the data points.
  4. **Evaluate the Model:** Find error in the model.
  5. **Make Predictions:** Use the regression equation to predict the value of the dependent variable for any given value of the independent variable.

Simple Regression Analysis: Methods of Evaluation

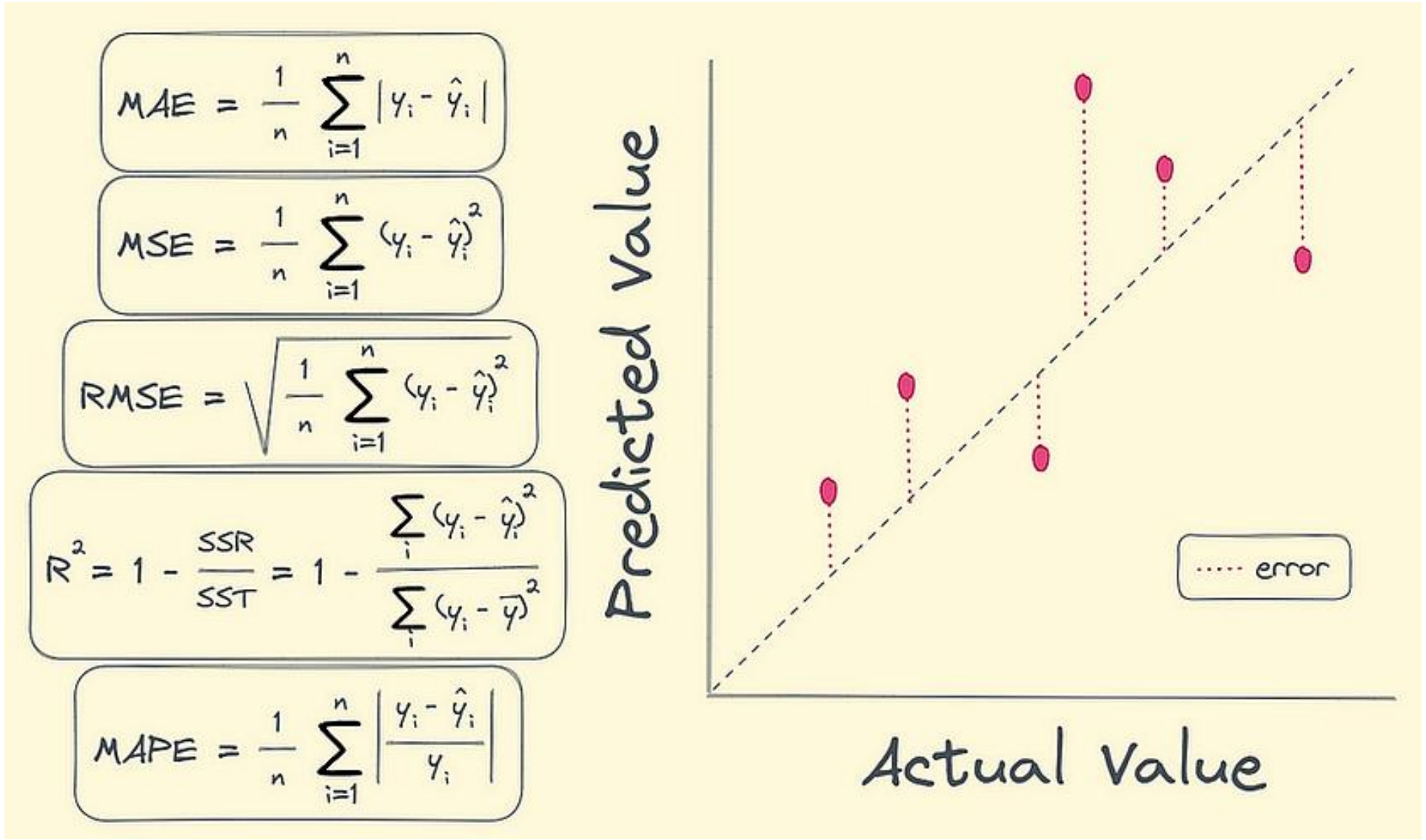
$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$



Simple Regression Analysis: Methods of Evaluation

. **Mean Absolute Error (MAE):** measures the average absolute difference between actual and predicted values. It treats all errors equally, regardless of their direction and provides results in the same unit as the target variable, making it easy to interpret.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Where:

n = total number of observations, y_i = actual value, $\hat{y}_i$ = predicted value

2. Mean Squared Error (MSE): calculates the average of squared differences between actual and predicted values. By squaring errors, it penalizes larger mistakes more strongly, making it sensitive to outliers.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Where:

n= number of observations, y_i = actual value, $\hat{y}_i$ = predicted value

Simple Regression Analysis: Methods of Evaluation

3. Root Mean Squared Error (RMSE): Root Mean Squared Error is the square root of MSE. It maintains the strong penalty for large errors while converting the result back to the original unit of the target variable, improving interpretability.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Where: n= number of observations, y_i = actual value, $\hat{y}_i$ = predicted value

4. Mean Absolute Percentage Error (MAPE): Mean Absolute Percentage Error expresses prediction error as a percentage of the actual values. It is intuitive and scale-independent but becomes unreliable when actual values are close to zero.

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

Where, n: total number of observations, y_i :actual value, $\hat{y}_i$:predicted value

Simple Regression Analysis: Methods of Evaluation

5. R-squared (R² Score): R-squared represents the proportion of variance in the target variable that is explained by the regression model. It provides insight into how well the model captures underlying data patterns.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}$$

Where: $\bar{y}$ =mean of actual values, Numerator = residual sum of squares, Denominator = total sum of squares

Multiple Linear Regression Model

- We consider the problem of regression when the study variable depends on more than one explanatory or independent variable, known as multiple linear regression.

- We can use it to find out which factor has the highest influence on the predicted output and how different variables are related to each other. Equation for multiple linear regression is:

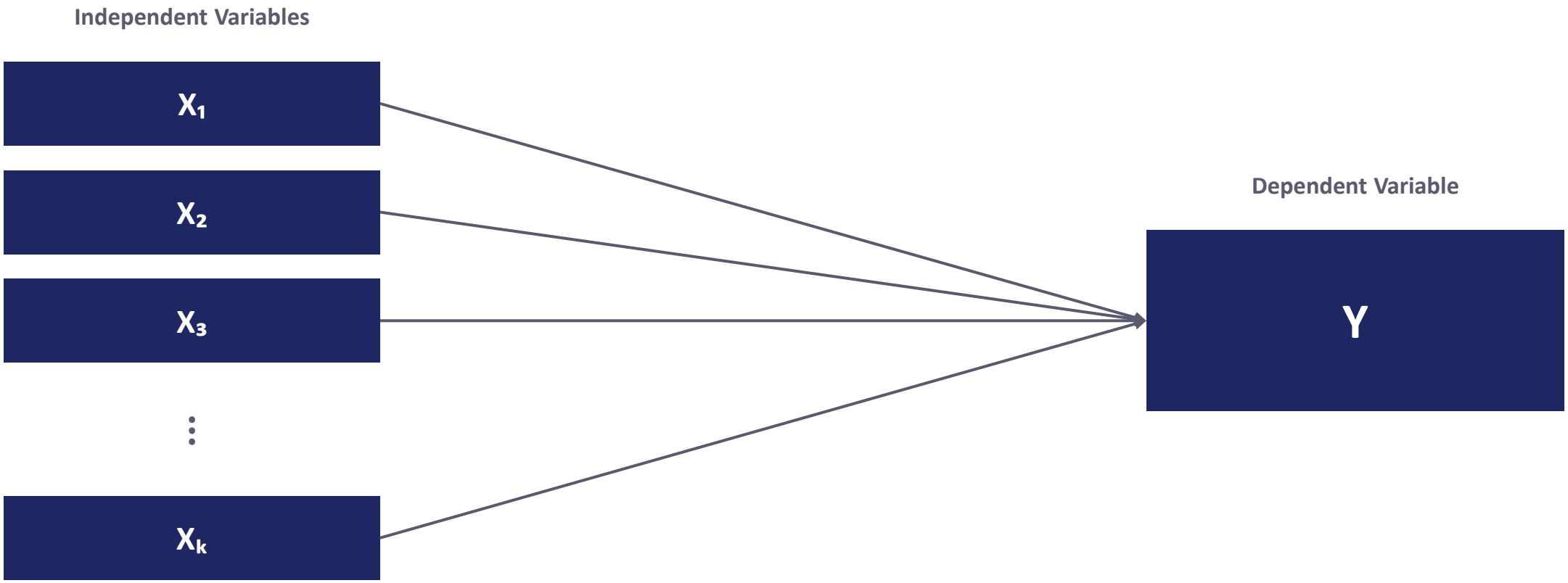
$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Where

- Y is the dependent variable
- $X_1, X_2, \cdots X_n$ are the independent variables
- β_0 is the intercept
- $\beta_1, \beta_2, \cdots \beta_n$ are the slopes

Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots + \beta_k X_k + \varepsilon$$



Each independent variable X_i contributes an effect β_i to the predicted value of Y, combined with an intercept β_0 and random error ε.

Multiple Linear Regression Model: Partial Regression Coeff.

- Equation for multiple linear regression is:

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Where

- *Y* is the dependent variable
- $X_1, X_2, \cdots X_n$ are the independent variables
- β_0 is the intercept
- $\beta_1, \beta_2, \cdots \beta_n$ are the slopes

→ Parital Regression Coefficient

Each independent variable X_i contributes an effect β_i to the predicted value of Y, combined with an intercept β_o and random error ε.

Multiple Linear Regression Model: Partial Regression Coeff.

Each **partial regression coefficient** (b_i) measures the **change in Y** for a **one-unit increase in X_i**, while keeping all other predictors constant.

Example

Suppose we have a model: $Y = 10 + 2X_1 + 3X_2$

- $b_1 = 2 \rightarrow$If X_1increases by 1 unit, Y increases by **2 units**, keeping X_2 constant.
- $b_2 = 3 \rightarrow$If X_2 increases by 1 unit, Y increases by **3 units**, keeping X_1 constant.

This isolates the effect of each variable, removing the influence of others.

Multiple Linear Regression Model: Partial Regression Coeff.

1. Regression Model

$$Y' = b_0 + b_1 X_1 + b_2 X_2 + e$$

2. Regression Coefficients (Slopes & Intercept)

Intercept b_0):

$$b_0 = \bar{Y} - b_1 \overline{X_1} - b_2 \overline{X_2}$$

Slope for X_1 (\beta_1):

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

Slope for $X_2 \beta_2$):

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

Regression Sum Calculations (Deviation Scores)

- **Sum of Squares for x_1:**

$$\sum x_1^2 = \sum X_1^2 - \frac{(\sum X_1)^2}{n}$$

- **Sum of Squares for x_2:**

$$\sum x_2^2 = \sum X_2^2 - \frac{(\sum X_2)^2}{n}$$

- **Sum of Products for $x_1 y$:**

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{n}$$

- **Sum of Products for $x_2 y$:**

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{n}$$

- **Sum of Products for $x_1 x_2$:**

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{n}$$

Multiple Linear Regression Model: Measuring goodness of fit.

Measuring Goodness of Fit: Coefficient of Determination (R^2): Let

- R : multiple correlation coefficient between $y\ and X_1, X_2, \dots, X_k$.
- Square of multiple correlation coefficient - R^2 is called as *coefficient of determination*.

Coefficient of Determination (R^2) :

The limits of R^2 are 0 and 1, i.e., $0 \leq R^2 \leq 1$

- $R^2 = 0$ indicates the poorest fit of the model.
- $R^2 = 1$ indicates the best fit of the model.
- $R^2 = 0.95$ indicates that 95% of the variation in y is explained by the explanatory variables.
- In simple words, the model is 95% good.
- Similarly, any other value of R^2 between 0 and 1 indicates the adequacy of fitted model

Multiple Linear Regression Model: Measuring goodness of fit.

The coefficient of determination (R^2) is expressed using following equation (2 variable):

$$R^2 = \frac{R^2 \sum_i y^2}{\sum y^2}$$

where

$$R^2 \sum_i y^2 = b_1 \sum x_1 y + b_2 \sum x_2 y$$

$$\sum y^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$$

Multiple Linear Regression Model: Measuring overfit.

- Standard R-squared measures the percentage of variation in your dependent variable explained by your predictors.
- It always increases when you add a new predictor, even if that predictor has no real relationship to the data.
- This causes overfitting, making a model look artificially better just because it has more variables
- Adjusted R-squared is a modified version of R-squared that penalizes the addition of useless variables in a multiple regression model.

Multiple Linear Regression Model: Measuring overfit.

Adjusted R-squared is a modified version of R-squared that adjusts for the number of predictors in the model. It gives a more accurate picture of how well your model is performing.

Adjusted R-Squared Formula:

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \cdot \frac{n - 1}{n - k - 1}$$

Where:

- n = number of observations
- k = number of independent variables (predictors)
- R^2 = the original R-squared

- It adjusts the score based on the number of predictors relative to the sample size.
- It only increases if a new variable improves the model more than expected by chance.
- It decreases if you add an irrelevant variable that does not add real value

Multiple Linear Regression Model: Overall Significance of Model.

- To test if the entire model is statistically significant, we use the F-test.
- Null & Alternative Hypotheses
  - $H_0: b_1 = b_2 = \cdots = b_n = 0$ →None of the predictors contribute significantly to Y.
  - H_A: At least one $b_i \neq 0$ →At least one predictor significantly explains Y.
- Formula for F-Statistic

$$F = \frac{MSR}{MSE} = \frac{Mean\ Square\ Regression}{Mean\ Square\ Error}$$

- Calculated F < Table F: Difference is insignificant → accept the null hypothesis.
- Calculated F ≥ Table F: Difference is significant → reject the null hypothesis.

Multiple Linear Regression Model: Overall Significance of Model.

Source of variation	Degrees of freedom (d.f.)	Sum of squares (SS)	Mean Square (MS)	F-ratio
Regression	$k = 2$	$R^2 \sum_i y^2$	$\frac{R^2 \sum_i y^2}{k}$	$\frac{(n-k-1)R^2}{k(1-R^2)}$
Error	$n-k-1$	$(1-R^2) \sum_i y^2$	$\frac{(1-R^2) \sum_i y^2}{n-k-1}$	–
Total	$n-1$	$\sum_i y^2$	–	–

- In case H_0 is accepted by F-test , there is no need to test the individual b's. Therefore, one must use the F-test first

Multiple Linear Regression Model: Testing for Individual Regression coefficients.

- When H_0 is rejected by F-test , we estimate coefficients $(b_1, b_2, \dots, b_n)$for each predictor variable. However, we need to check if each coefficient is **statistically significant** using a **t-test**.

1. Hypothesis for Individual Coefficients

   For each predictor X_i, we test:

   - Null Hypothesis (H_0): $b_i = 0$ (The predictor X_i has no effect on Y).
   - Alternative Hypothesis (H_A): $b_i \neq 0$(The predictor X_i significantly affects Y).

2. t-Test for Regression Coefficients

   To test if b_i is significantly different from zero, we calculate the t-statistic:

$$t = \frac{b_i}{SE(b_i)}$$

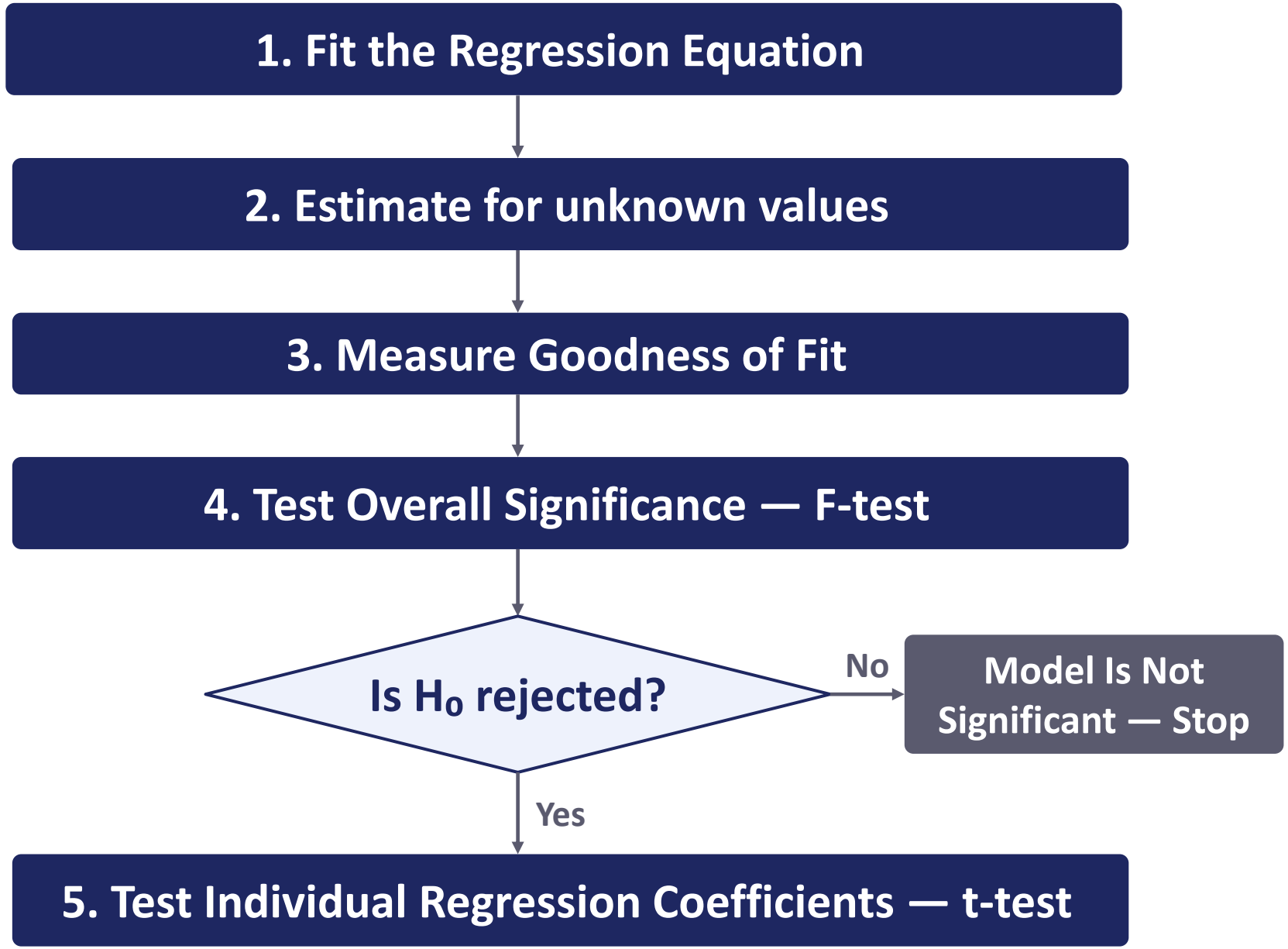
   Where:

   - b_i = Estimated regression coefficient
   - $SE(b_i)$ = Standard error of b_i, which measures the uncertainty in the coefficient estimate.

3. Interpretation

- If calculated t > tabulated t we reject $H_0 \rightarrow X_i$ is statistically significant.
- If calculated t < tabulated t we fail to reject $H_0 \rightarrow X_i$ might not be useful in the model.

Steps in Multiple Linear Regression Analysis



Steps in Multiple Linear Regression Analysis

Y	X_1	X_2
140	60	22
155	62	25
159	67	24
179	70	20
192	71	15
200	72	14
212	75	14
215	78	11

1. Fit the Regression Equation

1. Fit the Regression Equation.

1. Regression Model

$$Y' = b_0 + b_1 X_1 + b_2 X_2 + e$$

2. Regression Coefficients (Slopes & Intercept)

Intercept b_0):

$$b_0 = \bar{Y} - b_1 \overline{X_1} - b_2 \overline{X_2}$$

Slope for X_1 (\beta_1):

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

Slope for $X_2 \beta_2$):

$$b_2 = \frac{(\sum x_1^2)(\sum x_2 y) - (\sum x_1 x_2)(\sum x_1 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

Regression Sum Calculations (Deviation Scores)

- **Sum of Squares for x_1:**

$$\sum x_1^2 = \sum X_1^2 - \frac{(\sum X_1)^2}{n}$$

- **Sum of Squares for x_2:**

$$\sum x_2^2 = \sum X_2^2 - \frac{(\sum X_2)^2}{n}$$

- **Sum of Products for $x_1 y$:**

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{n}$$

- **Sum of Products for $x_2 y$:**

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{n}$$

- **Sum of Products for $x_1 x_2$:**

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{n}$$

1. Fit the Regression Equation.

1a. Calculate Sum of Squares

Y	X_1	X_2	Y^2	X_1^2	X_2^2	$X_1 \cdot X_2$	$X_1 \cdot Y$	$X_2 \cdot Y$
140	60	22	19,600	3,600	484	1,320	8,400	3,080
155	62	25	24,025	3,844	625	1,550	9,610	3,875
159	67	24	25,281	4,489	576	1,608	10,653	3,816
179	70	20	32,041	4,900	400	1,400	12,530	3,580
192	71	15	36,864	5,041	225	1,065	13,632	2,880
200	72	14	40,000	5,184	196	1,008	14,400	2,800
212	75	14	44,944	5,625	196	1,050	15,900	2,968
215	78	11	46,225	6,084	121	858	16,770	2,365
1,452	**555**	**145**	**268,980**	**38,767**	**2,823**	**9,859**	**101,895**	**25364**

1. Fit the Regression Equation.

1b. Calculate Regression Sums

- **Sum of Squares for x_1:**

$$\sum x_1^2 = \sum X_1^2 - \frac{(\sum X_1)^2}{n} == 38{,}767 - \frac{555^2}{8} = 38{,}767 - 38{,}503.125 = \mathbf{263.875}$$

- **Sum of Squares for x_2:**

$$\sum x_2^2 = \sum X_2^2 - \frac{(\sum X_2)^2}{n} = 2{,}823 - \frac{145^2}{8} = 2{,}823 - 2{,}628.125 = \mathbf{194.875}$$

- **Sum of Products for x_1y:**

$$\sum x_1 y = \sum X_1 Y - \frac{(\sum X_1)(\sum Y)}{n} = 101{,}895 - \frac{555 \times 1{,}452}{8} = 101{,}895 - 100{,}732.5 = \mathbf{1{,}162.5}$$

- **Sum of Products for x_2y:**

$$\sum x_2 y = \sum X_2 Y - \frac{(\sum X_2)(\sum Y)}{n} = 25{,}364 - \frac{145 \times 1{,}452}{8} = 25{,}364 - 26{,}317.5 = \mathbf{-953.5}$$

- **Sum of Products for x_1x_2:**

$$\sum x_1 x_2 = \sum X_1 X_2 - \frac{(\sum X_1)(\sum X_2)}{n} = 9{,}859 - \frac{555 \times 145}{8} = 9{,}859 - 10{,}059.375 = \mathbf{-200.375}$$

1. Fit the Regression Equation.

1c. Calculate Regression Coefficients (Slopes & Intercept)

Slope for X_1 β_1:

$$b_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2} = \frac{(194.875)(1162.5) - (-200.375)(-953.5)}{(263.875)(194.875) - (-200.375)^2} = \mathbf{3.1479}$$

Slope for $X_2\beta_2$:

$$b_2 = \frac{(263.875)(-683.5) - (-200.375)(1162.5)}{(263.875)(194.875) - (-200.375)^2} = -1.6561$$

Intercept b_0:

$$b_0 = \bar{Y} - b_1\overline{X_1} - b_2\overline{X_2}$$

$$With\ means\ \bar{Y} = 1452/8 = 181.5, \overline{X_1} = 555/8 = 69.375, \overline{X_2} = 145/8 = 18.125:$$

$$b_0 = \bar{Y} - b_1\overline{X_1} - b_2\overline{X_2} = 181.5 - (\mathbf{3.1479})(69.375) - (-1.6561)(18.125) = -6.8675$$

1. Fit the Regression Equation.

1d. Regression Model

$$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2 + e$$

$$\hat{Y} = -6.8675 + \mathbf{3.1479} X_1 - 1.6561 X_2$$

Interpretation

Holding X_2 constant, each 1-unit increase in X_1 raises predicted Y by ≈ 3.1479 units.

Holding X_1 constant, each 1-unit decrease in X_2 raises predicted Y by ≈ 1.6561 units.

2. Estimate for unknown values

2. Estimate for unknown values

Regression Model : $\hat{Y} = -6.8675 + \mathbf{3.1479}X_1 - 1.6561X_2$

Predict $\hat{Y}$ for X_1= 74 & X_2=14.

$$\hat{Y} = -6.8675 + \mathbf{3.1479}(74) - 1.6561(14)$$

$$\boxed{\hat{Y} \approx 202.893}$$

Interpretation: For an individual with X_1 = 74 and X_2 = 14, the model predicts Y ≈ 202.893

3. Measure Goodness of Fit

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The coefficient of determination (R^2) is expressed using following equation (2 variable):

$$\mathrm{R}^2 = \frac{\mathrm{R}^2 \sum_{\mathrm{i}} \mathrm{y}^2}{\sum \mathrm{y}^2}$$

where

$$\mathrm{R}^2 \sum_{\mathrm{i}} \mathrm{y}^2 = b_1 \sum x_1 y + b_2 \sum x_2 y \quad \text{and} \quad \sum \mathrm{y}^2 = \sum \mathrm{Y}^2 - \frac{(\sum \mathrm{Y})^2}{\mathrm{n}}$$

Step 1 — Total sum of squares, Σy²

$$\sum y^2 = \sum Y^2 - \frac{(\sum Y)^2}{n} = 268{,}980 - \frac{1{,}452^2}{8} = 268{,}980 - 263{,}538 = \mathbf{5{,}442}$$

Step 2 — Explained sum of squares

$$\mathrm{R}^2 \sum_{\mathrm{i}} \mathrm{y}^2 = b_1 \sum x_1 y + b_2 \sum x_2 y = (3.1479)(1{,}162.5) + (-1.6561)(-953.5) = 3{,}659.43 + 1{,}579.13$$

$$= \mathbf{5{,}238.56}$$

Step 3 — Coefficient of determination

$$R^2 = \frac{5{,}238.56}{5{,}442} = \boxed{0.9626; (\approx 96.3\backslash\%)}$$

Interpretation: About 96.3% of the variation in Y is explained jointly by X_1 and X_2 — a very strong fit.

3. Measure Goodness of Fit

Step 1 — Total sum of squares, Σy²

$$\sum y^2 = \sum Y^2 - \frac{(\sum Y)^2}{n} = 268{,}980 - \frac{1{,}452^2}{8} = 268{,}980 - 263{,}538 = \mathbf{5{,}442}$$

Step 2 — Explained sum of squares

$$R^{2}\sum y^2 = b_1 \sum x_1 y + b_2 \sum x_2 y = (3.1479)(1{,}162.5) + (-1.6561)(-953.5) = 3{,}659.43 + 1{,}579.13 = \mathbf{5{,}238.56}$$

Step 3 — Coefficient of determination

$$R^2 = \frac{5{,}238.56}{5{,}442} = \boxed{0.9626; (\approx 96.3\backslash\%)}$$

Interpretation: About 96.3% of the variation in Y is explained jointly by X_1 and X_2 — a very strong fit.

4. Test Overall Significance — F-test

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Source of variation	Degrees of freedom (d.f.)	Sum of squares (SS)	Mean Square (MS)	F-ratio
Regression	$k = 2$	$R^2 \sum_i y^2$	$\frac{R^2 \sum_i y^2}{k}$	$\frac{(n-k-1)R^2}{k(1-R^2)}$
Error	$n-k-1$	$(1-R^2) \sum_i y^2$	$\frac{(1-R^2) \sum_i y^2}{n-k-1}$	—
Total	$n-1$	$\sum_i y^2$	—	—

4. Test Overall Significance — F-test

Source of variation	d.f.	Sum of Squares (SS)	Mean Square (MS)	F-ratio
Regression	$k = 2$	$R^2 \cdot \Sigma y^2 = (0.9626)(5{,}442) = 5{,}238.56$	$\frac{R^2 \sum_i y^2}{k} = 5{,}238.56 / 2 = 2{,}619.28$	$\frac{(n-k-1)R^2}{k(1-R^2)} = \frac{(5)(0.9626)}{(2)(0.0374)} = \frac{4.813}{0.0748} \approx \mathbf{64.35}$
Error	$n - k - 1 = 8 - 2 - 1 = 5$	$(1 - R^2) \cdot \Sigma y^2 = (0.0374)(5{,}442) = 203.44$	SS_error / (n−k−1) = 203.44 / 5 = 40.69	—
Total	$n - 1 = 8 - 1 = 7$	$\Sigma y^2 = 5{,}442.00$	—	—

Conclusion: Compare F = 64.37 against the critical value $F_{0.05}(2, 5) \approx 5.79$.

Since 64.37 ≫ 5.79, we **reject H_0** — the regression model is overall statistically significant, meaning X_1 and X_2 jointly explain a significant portion of the variation in Y.

This means we'd proceed to test the individual coefficients (t-test) next,.

5. Test Individual Regression Coefficients — t-test

5. Test Individual Regression Coefficients — t-test

1. Hypothesis for Individual Coefficients

   For each predictor X_i, we test:

   - Null Hypothesis (H_0): $b_i = 0$ (The predictor X_i has no effect on Y).
   - Alternative Hypothesis (H_A): $b_i \neq 0$(The predictor X_i significantly affects Y).

2. t-Test for Regression Coefficients

   To test if b_i is significantly different from zero, we calculate the t-statistic:

$$t = \frac{b_i}{SE(b_i)}$$

   Where:

   - b_i = Estimated regression coefficient
   - $SE(b_i)$ = Standard error of b_i, which measures the uncertainty in the coefficient estimate.

3. Interpretation

- If calculated t > tabulated t we reject $H_0 \rightarrow X_i$ is statistically significant.
- If calculated t < tabulated t we fail to reject $H_0 \rightarrow X_i$ might not be useful in the model.

5. Test Individual Regression Coefficients — t-test

$$\hat{Y} = -6.8675 + \mathbf{3.1479}X_1 - 1.6561X_2$$

	X_2	Y_i	$\hat{Y}_i$	RSS: $\Sigma(Y_i-\hat{Y}_i)^2$	S_{b_1}: $\Sigma(X_1-\bar{X}_1)^2$	S_{b_2}: $\Sigma(X_2-\bar{X}_2)^2$
60	22	140	145.57	31.05	87.98	14.98
62	25	155	146.90	65.61	54.46	47.20
67	24	159	164.30	28.04	5.66	34.46
70	20	179	180.36	1.86	0.38	3.50
71	15	192	191.79	0.04	2.62	9.80
72	14	200	196.60	11.59	6.86	17.06
75	14	212	206.04	35.53	31.58	17.06
78	11	215	220.45	29.72	74.30	50.84
$\bar{X}_1$ = 69.38	**$\bar{X}_2$ = 18.13**	**$\bar{Y}$ = 181.5**		**Σ = 203.44**	**Σ = 263.88**	**Σ = 194.88**

Correlation coefficient r_{12}

$$r_{12} = \frac{\sum x_1 x_2}{\sqrt{(\sum x_1^2)(\sum x_2^2)}} = \frac{-200.375}{\sqrt{(263.875)(194.875)}} = -0.88$$

Degree of freedom

$$\text{d.f.} = \text{n} - \text{k} - 1 = 8 - 2 - 1 = 5$$

N=number of observation

k= independent variable

5. Test Individual Regression Coefficients — t-test

1. Hypothesis for Individual Coefficients b_1
   - Null Hypothesis (H_0): $b_1 = 0$ (The predictor X_1 has no effect on Y).
   - Alternative Hypothesis (H_A): $b_1 \neq 0$(The predictor X_1 significantly affects Y).
2. t-Test for Regression Coefficients

   To test if b_1 is significantly different from zero, we calculate the t-statistic:

$$t = \frac{b_1}{SE(b_1)}$$

$$SE(b_1) = \frac{\sqrt{\frac{\sum(y_i - \hat{y})^2}{n-k-1}}}{\sqrt{\sum(x_1 - \overline{x_1})^2 \times (1 - r_{12}^2)}} = \frac{\sqrt{\frac{203.44}{5}}}{\sqrt{263.88 \times (1 - (-0.88)^2)}} = 0.838$$

$$t = \frac{3.1479}{0.838} = 3.756$$

3. Interpretation
- If calculated t (3.756) > tabulated $t_{(0.05,5)} = 2.571$ we reject $H_0 \rightarrow X_1$ is statistically significant.

5. Test Individual Regression Coefficients — t-test

1. Hypothesis for Individual Coefficients b_2
   - Null Hypothesis (H_0): $b_2 = 0$ (The predictor X_2 has no effect on Y).
   - Alternative Hypothesis (H_A): $b_2 \neq 0$(The predictor X_2 significantly affects Y).
2. t-Test for Regression Coefficients

   To test if b_2 is significantly different from zero, we calculate the t-statistic:

$$t = \frac{b_2}{SE(b_2)}$$

$$SE(b_2) = \frac{\sqrt{\frac{\sum(y_i - \hat{y})^2}{n-k-1}}}{\sqrt{\sum(x_2 - \overline{x_2})^2 \times (1 - r_{12}^2)}} = \frac{\sqrt{\frac{203.44}{5}}}{\sqrt{263.88 \times (1 - (-0.88)^2)}} = 0.838$$

$$t = \frac{3.1479}{0.838} = 3.756$$

3. Interpretation
   - If calculated t (-1.72) < tabulated $t_{(0.05,5)} = 2.571$ we accept $H_0 \rightarrow X_2$ is not statistically significant.